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B.S. in Mechanical Engineering and Physics, August 2025–May 2030.

PsiSat

[Physics proposal](#)

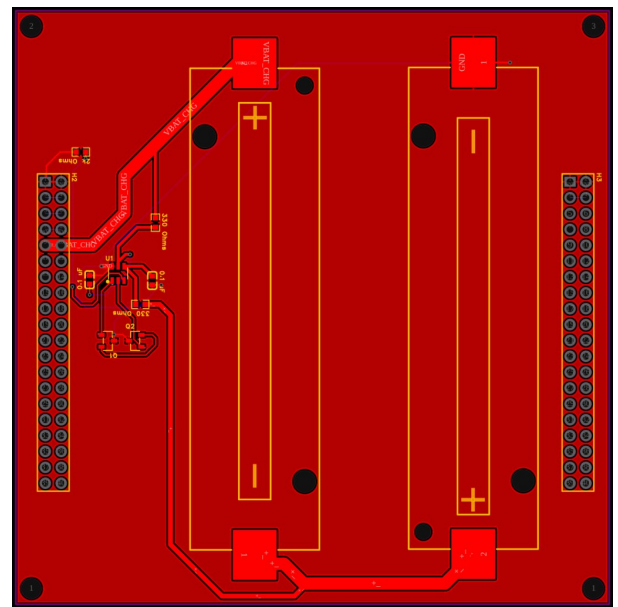
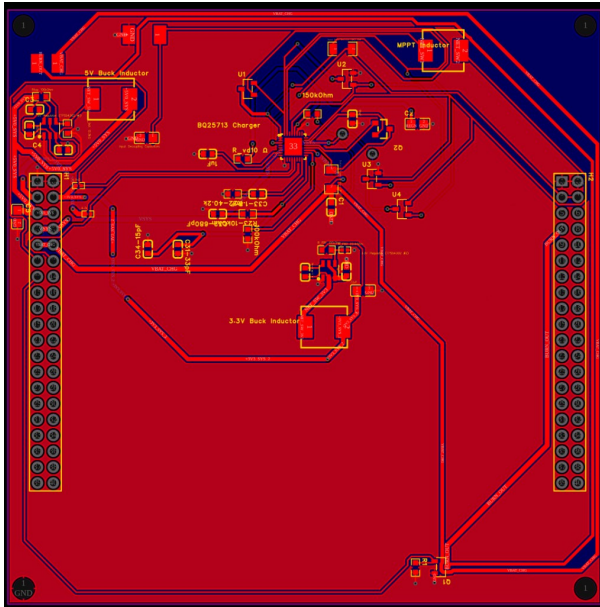
I started PsiSat at Stony Brook. It is also called LEO-GRM. The goal is a 1U CubeSat, a box about 10 cm on a side and under 1.33 kg, that notices short flashes of gamma rays from orbit.

A gamma ray hits a small crystal. The crystal gives off light near 550 nm. A light sensor reads that flash, and a small computer records the time. The work is a proposal for NASA's CubeSat Launch Initiative. It has not flown.

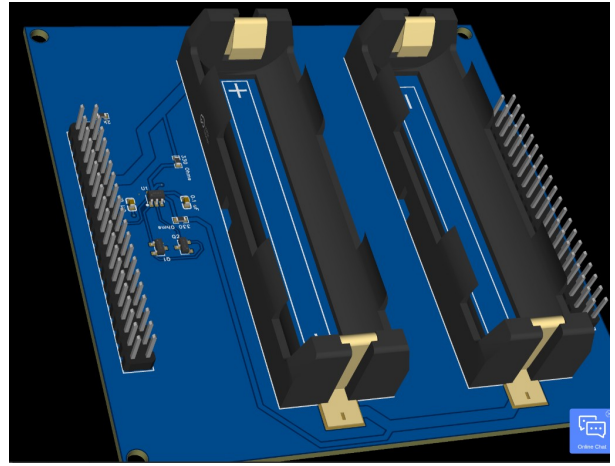
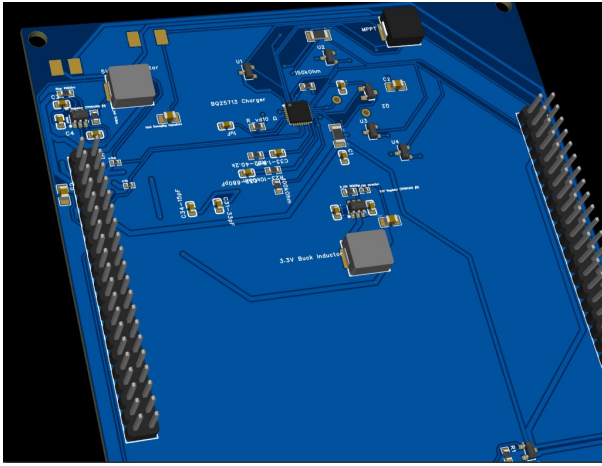
The electronics are planned as six stacked boards. Four of those boards are still being drawn. The two shown here are finished enough to show.

Stage 4 is the power board. It charges the pack, tracks power from the solar cells, and makes the 5 V and 3.3 V supplies.

Stage 5 is the battery board. It holds the cells.



Left: Stage 4 layout. Right: Stage 5 layout, with the cell tracks.



Left: Stage 4, the board in 3D. Right: Stage 5, the battery holders in 3D.

3D-printed jet

This airplane was drawn in Fusion 360 and printed in parts. Two 80 mm electric ducted fans sit side by side at the rear.

The wing halves are printed separately. They close over a battery bay in the fuselage.

The design target is more than 70 km/h. Work is still going on tilting the thrust, and on flight and landing without a pilot. The model is on the website.



CAD model of the airframe, fans, and tail.



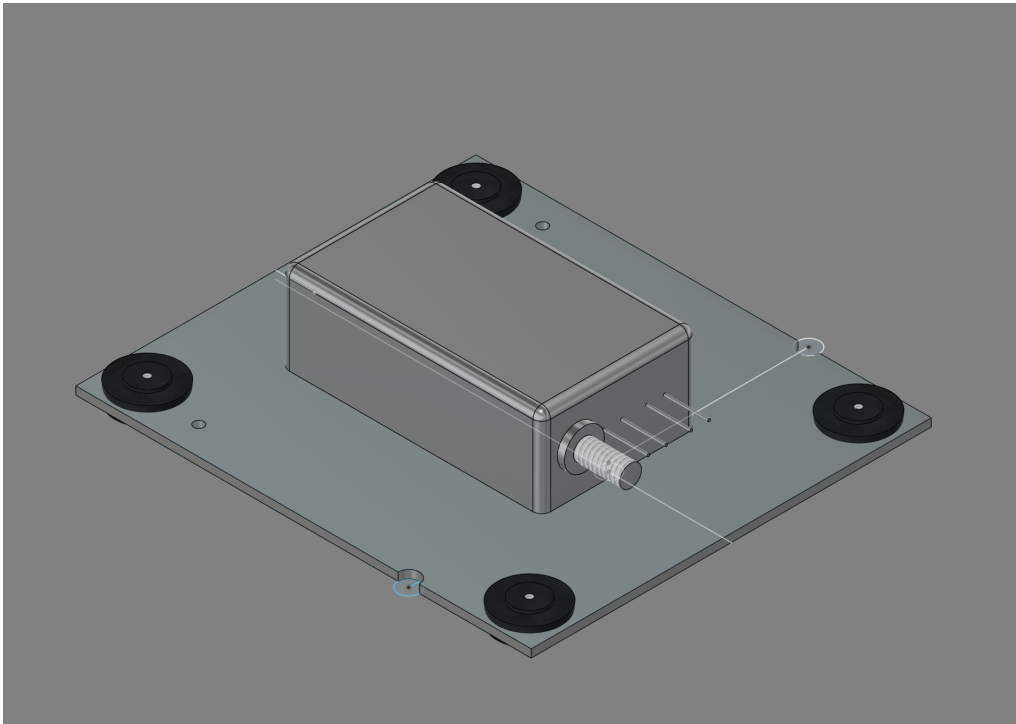
Left: printed wing panels and the battery bay. Right: the twin fan ducts and the tail, during assembly.

Mechanical engineering intern at East End Assemblies, Long Island.

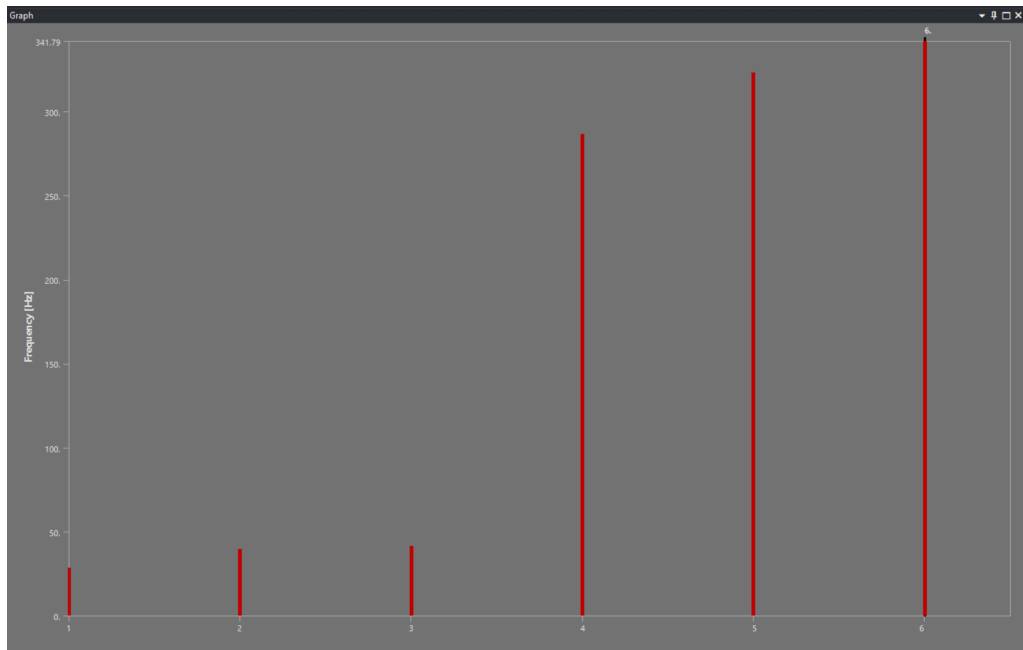
The July 2026 study is a 10 MHz crystal oscillator, Wenzel 501-27521-11, on a plate with four J-3112-2 isolators. I ran a modal analysis in Ansys Mechanical and read the first six natural frequencies.

The low modes sit near 30–50 Hz. If the bench shakes in that range, the mounts amplify the motion. Isolation starts to help only once the disturbance is above about 70 Hz. Above a few hundred hertz the mounts stiffen and stop isolating.

The point of the study was to say where the mounts help, and where they do not, before the oscillator is bolted down.



Oscillator plate and the four isolators.



Modal results in Ansys. Modes 1–6.

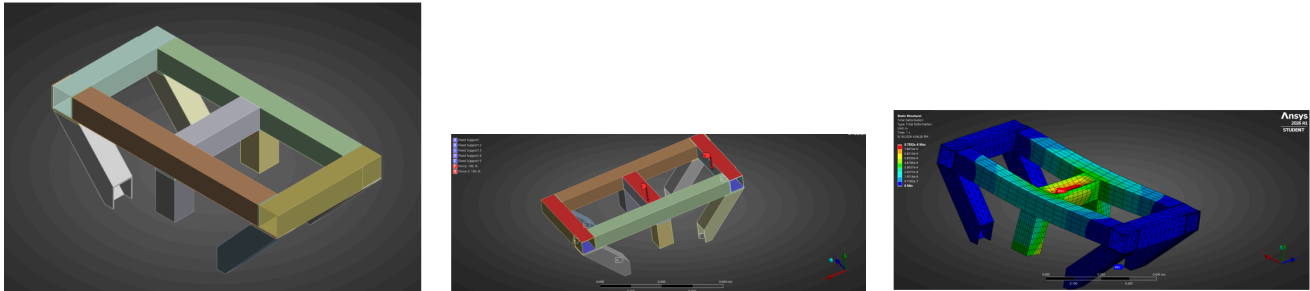
FEA lead, Stony Brook Robotics

September 2025–present. I design the URC rover frame in SolidWorks, then check it before it is built.

Ansys Mechanical is the usual study: static loads on the frame. LS-DYNA is for the short, hard impact cases. Running the frame through several load cases cut 6 kg from the rover.

One of those studies, 16 September 2026, is the top frame. The welded joints were modeled as 5 mm pieces tying the tubes together, all in aluminum 6061-T6. Three loads sit on the members: 100 N and two forces of 150 N. The frame is held on a few faces so it cannot float. The mesh was left at the default for this pass.

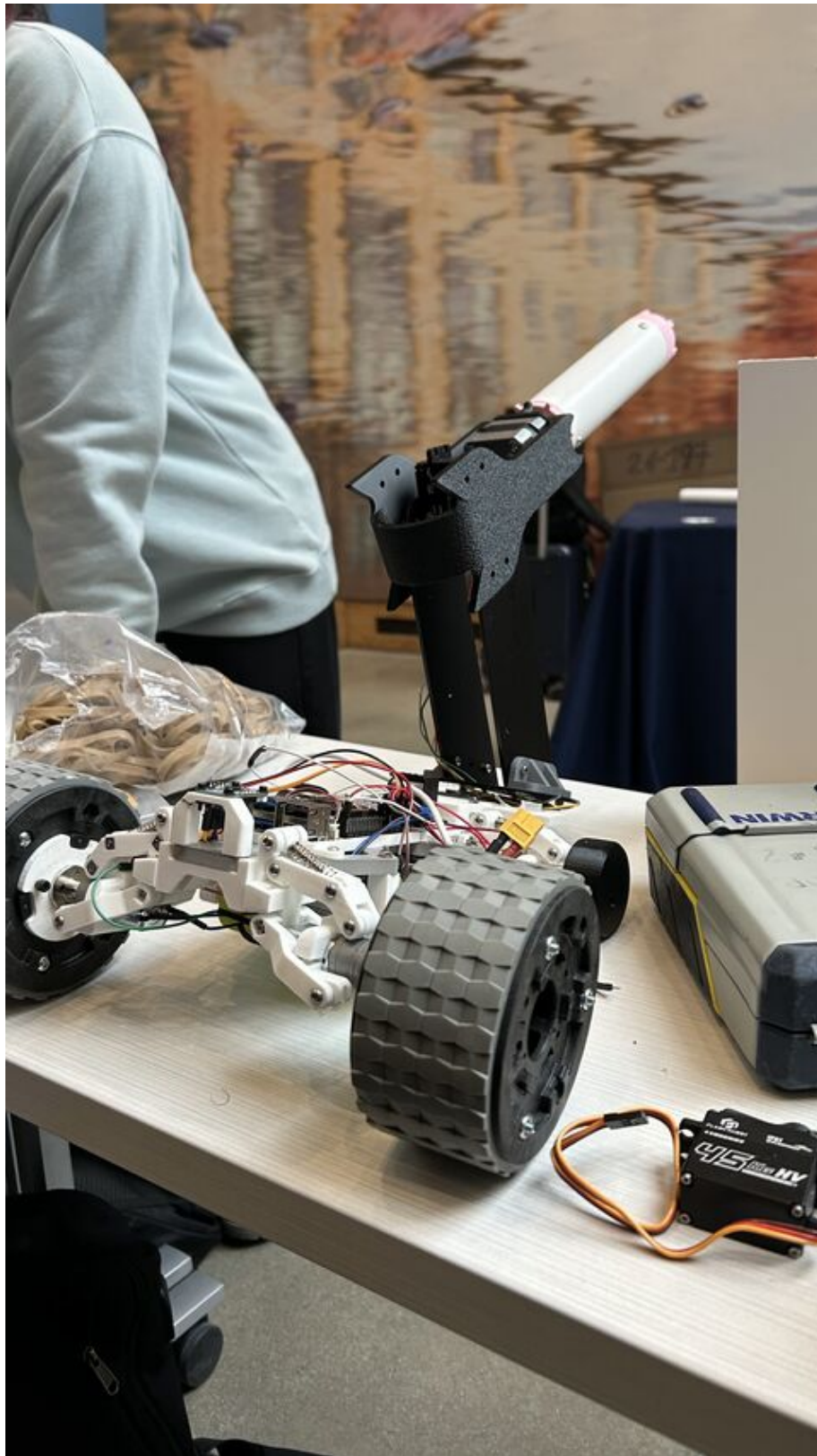
The deformation plot peaks on the middle member. That is a layout problem, not a bad alloy. The next pass adds supports there, trusses or L-bars with a fillet instead of a sharp corner, and checks a 2 by 2 inch tube size.



Left: the top frame. Center: the three loads and the fixed faces. Right: total deformation. The middle member is the hot spot.

A wheeled robot with a small turret. The turret aims and fires short 3D-printed cylinders.

A camera and face recognition decide when a person in front of the robot is treated as an intruder. The mechanical job was to put the launcher, the aiming hardware, and the wiring on a rover that can still roll.



The rover and turret on the bench.

Other roles

Teaching assistant, AMS 261. Multivariable Calculus, Stony Brook.

Autodesk ambassador, Stony Brook. January 2026–present. I hosted a workshop on Fusion 360, analysis, and rendering.

Tools

CAD: SolidWorks, Fusion 360, Inventor, Onshape, Solid Edge.

Analysis: Ansys Mechanical, Ansys Fluent, Workbench, LS-DYNA, OpenFOAM, SimScale.

Software: C++, KiCad, PyTorch, Qiskit, OpenRocket, L^AT_EX.